MACHINE LEARNING BASED WEATHER PREDICTION

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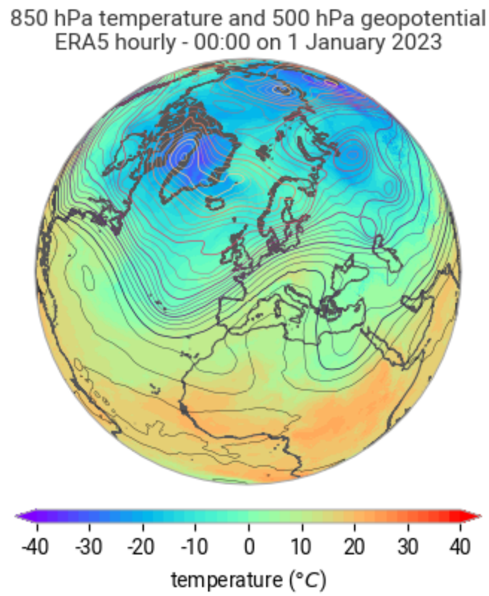
Introduction

Why weather?

The weather, on any given day, is important for almost every industry and activity a person can think of. From simple recreational activities like skiing all the way to the hyper complex organization of global shipping lanes, weather plays an integral role. Weather can reach us even in places where it physically cannot travel. Wind speed and solar predictions affect prices on the stock market based on renewable energy prediction. Even infrastructure as wide and ubiquitous as a state's electricity grid can be brought to a grinding halt by the power of heat domes or hurricanes.



1. Beautiful chart depicting temperature, geopotential, and isobars over the Arctic, Europe, and North Africa.

Precisely because of the wide reaching impacts of weather, large quantities of accessible data from many sources exist for use in modeling and research. This huge amount of information makes the topic ideal for an aspiring data scientist. Many large models from around the globe are in production, with many more under development. These studies can be used as inspiration and guides for those of us beginning to dip our toes in the field.

The impetus for this project comes from multiple places. As a lifelong surfer, skier, and general outdoor enthusiast, the weather has always played an important role in my favorite activities. I've been poring over marine forecasts to predict swell at my favorite beaches since I was 12. Sometime later, when I found myself exploring alpinism and backcountry skiing, I added avalanche reports and mountain weather to the list of charts that needed understanding. Now, after nearly two decades of this self-forecasting, I've found myself a part of an OSU CEOAS lab where the understanding of how swell might propagate to our field site in the Caribbean is of great importance for data collection.

The true prediction of weather is beyond the scope of this course and probably an undergraduate degree in general. However, this seems to be a perfect opportunity to explore atmospheric data in a structured way. A primary goal for this project is to gain some understanding of how processes in the atmosphere are measured, as well as how non-weather specific events can be related to the weather in a way that is apparent through the data. The latter of these goals is most obviously illustrated my work with the urban air quality dataset at the end of this report.

Gathering Data

Direct Data Download

Urban Air Quality Data

https://www.kaggle.com/datasets/abdullah0a/urban-air-quality-and-health-impact-dataset

This dataset focused on urban air quality provides real measurements of 10 different cities' AQI for the month of September in 2024. Columns span the gamut from meteorological data such as windspeed, precipitation, and temperature; to human dimensions such as the day of the week, health risk score, and whether it's the weekend or not.

```
datetime,datetimeEpoch,tempmax,tempmin,temp,feelslikemax,feelslikemin,feelslike,dew,humidity,precip,precipprob,precipcover,preciptype,snow,snowdepth,windgust,windspeed,
2024-09-07,1725692400.0,106.1,91.0,98.5,104.0,88.1,95.9,51.5,21.0,0.0,5.0,0.0,,0.0,0.0,26.3,13.7,107.3,1009.2,25.0,10.0,261.4,22.5,9.0,10.0,06:06:50,1725714410.0,18:44:5
2024-09-08,1725778800.0,103.9,87.0,95.4,100.5,84.7,92.3,48.7,21.5,0.0,3.0,0.0,,0.0,0.0,20.8,12.8,101.5,1008.8,13.5,10.1,293.3,25.2,9.0,10.0,06:07:30,1725800850.0,18:43:3
2024-09-09,1725865200.0,105.0,83.9,94.7,99.9,81.6,90.6,41.7,16.9,0.0,0.0,0.0,,0.0,0.0,18.3,10.3,90.8,1009.4,6.2,10.1,327.0,28.2,9.0,10.0,06:08:10,1725887290.0,18:42:10,
2024-09-10,1725951600.0,106.1,81.2,93.9,100.6,79.5,89.8,39.1,15.7,0.012,0.0,4.17,['rain'],0.0,0.0,10.5,5.4,130.1,1006.8,4.9,12.5,276.8,24.0,9.0,10.0,06:08:49,1725973729
2024-09-11,1726038000.0,106.1,82.1,94.0,101.0,80.0,90.0,40.1,15.9,0.008,0.0,4.17,['rain'],0.0,0.0,15.9,8.1,201.6,1001.8,5.7,15.0,274.9,23.7,9.0,10.0,06:09:29,1726060169
2024-09-12,1726124400.0,103.0,81.6,91.5,98.1,79.9,87.9,41.6,18.4,0.0,0.0,0.0,,0.0,0.0,21.5,10.7,201.9,1000.8,4.0,15.0,272.7,23.6,9.0,10.0,06:10:09,1726146609.0,18:38:04
2024-09-13,1726210800.0,101.2,77.6,89.3,96.9,77.6,86.4,42.6,20.4,0.0,0.0,0.0,,0.0,0.0,14.3,7.2,192.9,1006.3,7.2,15.0,269.5,23.1,9.0,10.0,06:10:49,1726233049.0,18:36:41,
2024-09-14,1726297200.0,99.4,78.2,89.1,96.1,78.2,86.8,48.4,25.3,0.004,6.0,4.17,['rain'],0.0,0.0,12.1,5.4,224.8,1008.7,9.6,15.0,262.5,22.6,9.0,10.0,06:11:29,1726319489.0
2024-09-15,1726383600.0,100.1,82.1,92.0,100.6,81.5,92.3,60.3,35.8,0.012,12.9,4.17,['rain'],0.0,0.0,19.0,11.6,254.3,1007.3,1.8,14.0,259.9,22.3,8.0,30.0,06:12:09,17264059
```

Curated Rain Occurrence Data

https://www.kaggle.com/datasets/zeeshier/weather-forecast-dataset

This dataset is focused on predicting whether or not rain occurred on a given day, based on a variety of factors such as cloud cover, pressure, or humidity. The majority of the data is quantitative with just a single qualitative column of "rain" or "no rain" which is to be used as labels for ML.

```
Temperature,Humidity,Wind_Speed,Cloud_Cover,Pressure,Rain
23.720337598183118,89.59264065174611,7.335604391040214,50.501693832913155,1032.378758690279,rain
27.879734159310487,46.48970403534824,5.952483593282764,4.990052927536981,992.6141895121403,no rain
25.069084401791095,83.07284289257146,1.3719918180799207,14.855783939243427,1007.2316201172738,no rain
23.622079574922424,74.36775792086564,7.050550632784658,67.25528206034686,982.6320127095369,rain
20.591369983472617,96.85882241307947,4.6439209259534975,47.67644427890656,980.8251417426507,no rain
26.147352826666403,48.21726041042326,15.258546760433369,59.76627863227763,1049.738751014439,no rain
20.939680281567313,40.79944443606025,2.2325659971252265,45.827508421021676,1014.1737660638806,no rain
32.29432501955199,51.848471069719,2.873620839084101,92.5514972525134,1006.0417334780344,no rain
34.09156901252573,48.05711447385431,5.570206017562278,82.52487276907972,993.7320466194785,no rain
19.58603797064444,82.9782932331459,5.7605365867061025,98.0144498972855,1036.5034571561982,rain
29.793125952066614,81.31765126927587,16.92609873392226,93.92329417135517,1029.4026903671104,no rain
23.22237299382261,76.87794305987336,15.825673129009745,72.86978986682412,980.1089338550743,rain
24.201114027348307,45.14653801209415,11.572712664104507,5.25304210213815,1033.9858671128165,no rain
```

Data Cleaning

Urban AQI Data

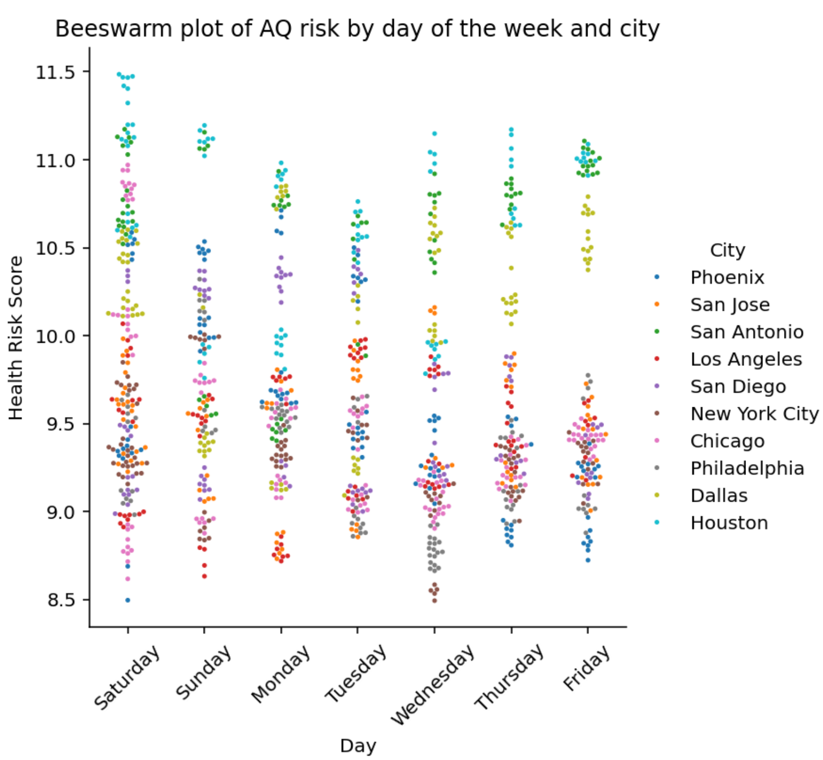
Before Cleaning

Example Data

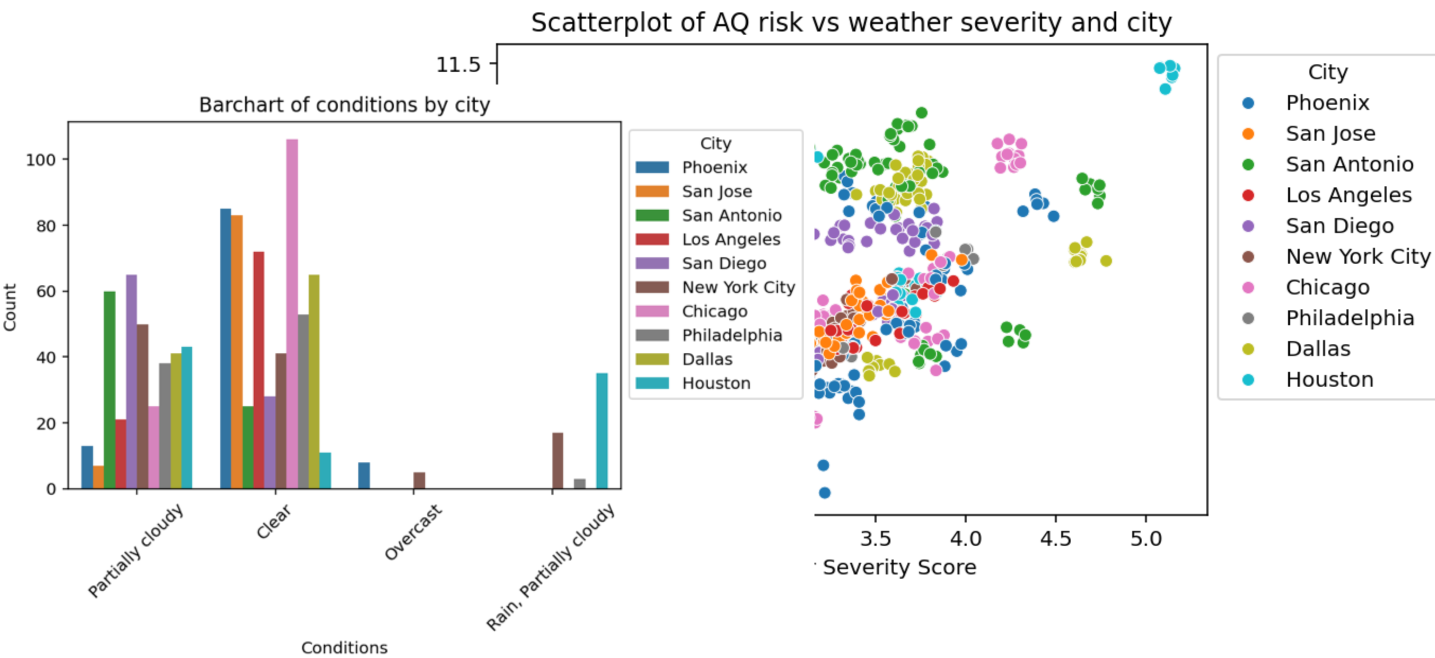
```
        datetime  datetimeEpoch  ...  Is_Weekend  Health_Risk_Score
0     2024-09-07   1.725692e+09  ...        True          10.522170
1     2024-09-08   1.725779e+09  ...        True          10.062332
2     2024-09-09   1.725865e+09  ...       False           9.673387
3     2024-09-10   1.725952e+09  ...       False           9.411519
4     2024-09-11   1.726038e+09  ...       False           9.515179
..           ...            ...  ...         ...                ...
995   2024-09-18   1.726633e+09  ...       False           8.750142
996   2024-09-17   1.726550e+09  ...       False           9.118198
997   2024-09-12   1.726122e+09  ...       False           9.880093
998   2024-09-14   1.726284e+09  ...        True           9.561602
999   2024-09-18   1.726618e+09  ...       False          10.978044

[1000 rows x 46 columns]
```

Visualizing Variables



Initial EDA indicates an interesting trend between air pollution and the day of the week. Across the board, the health risk score seems to be lower for most cities on the weekdays. Air quality decreases most on Saturdays in general. This plot does not reveal any outliers or erroneous data points, but it's quite informative for the structure of the data.



Further exploration of air quality, while taking the city in question into account, revealed an interesting linear trend in conjunction with the severity of the day's weather. The scatter plot shows two fairly strong clumps of data. Nearly all the cities in the upper section are located Texas! Only Houston exhibits data approaching an outlier level of significance. More investigation is needed to be sure.

A bar chart of general sky conditions and the cities they occur in most often is shown to the left. Much this plot is fairly sparse, with most cities only reporting either Partially Cloudy or Clear. This could indicate that the data is not very representative of the cities in question. If all these locations always have this nice of weather in September, how can the analysis derive insight on the impact of poor weather on the AQI?

After Cleaning

The Urban AQI dataset was downloaded as a .csv. Initial EDA seemed to indicate that the dataset is quite clean already, as is often the case when data is gathered from Kaggle. After some simple cleaning operations, the final, ready-for-analysis data will be stored as a new .csv.

A majority of the data types for this dataset were read in correctly by Pandas. The "datetime" variable was changed to a datetime type. "City", "conditions", and "Day_of_Week" were all converted to categories. Lastly, "Month" was converted to an integer.

```
datetime              object
tempmax              float64
tempmin              float64
temp                 float64
feelslikemax         float64
feelslikemin         float64
feelslike            float64
dew                  float64
humidity             float64
precip               float64
precipprob           float64
precipcover          float64
windgust             float64
windspeed            float64
winddir              float64
pressure             float64
cloudcover           float64
visibility           float64
solarradiation       float64
solarenergy          float64
uvindex              float64
severerisk           float64
sunrise               object
sunset                object
conditions            object
City                  object
Temp_Range           float64
Heat_Index           float64
Severity_Score       float64
Month                float64
Day_of_Week           object
Is_Weekend              bool
Health_Risk_Score    float64
dtype: object
```

```
datetime             datetime64[ns]
tempmax                     float64
tempmin                     float64
temp                        float64
feelslikemax                float64
feelslikemin                float64
feelslike                   float64
dew                         float64
humidity                    float64
precip                      float64
precipprob                  float64
precipcover                 float64
windgust                    float64
windspeed                   float64
winddir                     float64
pressure                    float64
cloudcover                  float64
visibility                  float64
solarradiation              float64
solarenergy                 float64
uvindex                     float64
severerisk                  float64
sunrise                      object
sunset                       object
conditions                 category
City                       category
Temp_Range                  float64
Heat_Index                  float64
Severity_Score              float64
Month                         int64
Day_of_Week                category
Is_Weekend                     bool
Health_Risk_Score           float64
dtype: object
```

Once the columns, 'datetimeEpoch', 'sunriseEpoch', 'sunsetEpoch', 'moonphase', 'description', 'icon', 'stations', 'source', 'Condition_Code', 'preciptype', 'snow', 'snowdepth', 'Season' were dropped, not a single null value remained in the dataset. These columns were dropped because they either contained all 0's, as in the case of the "snowdepth" variable, or because they seemed unnecessary for analysis.

Working with all 25 quantitative columns to discover outliers was a challenge. In the end, the 1st and 3rd quartiles, and the interquartile range for each column was calculated. From there, a Boolean mask using the quartile outlier formula was used to uncover columns with at least one record that contained a possible outlier.

$$low\ outliers = Q1 - 1.5 * IQR$$
$$high\ outliers = Q3 + 1.5 * IQR$$

Using a loop, each of the suspicious variables was inspected with a boxplot. From this exploration, only two variables had outliers of any significance: "dew" and "windspeed". Both records were cleaned using a Boolean mask.

Rain Occurrence

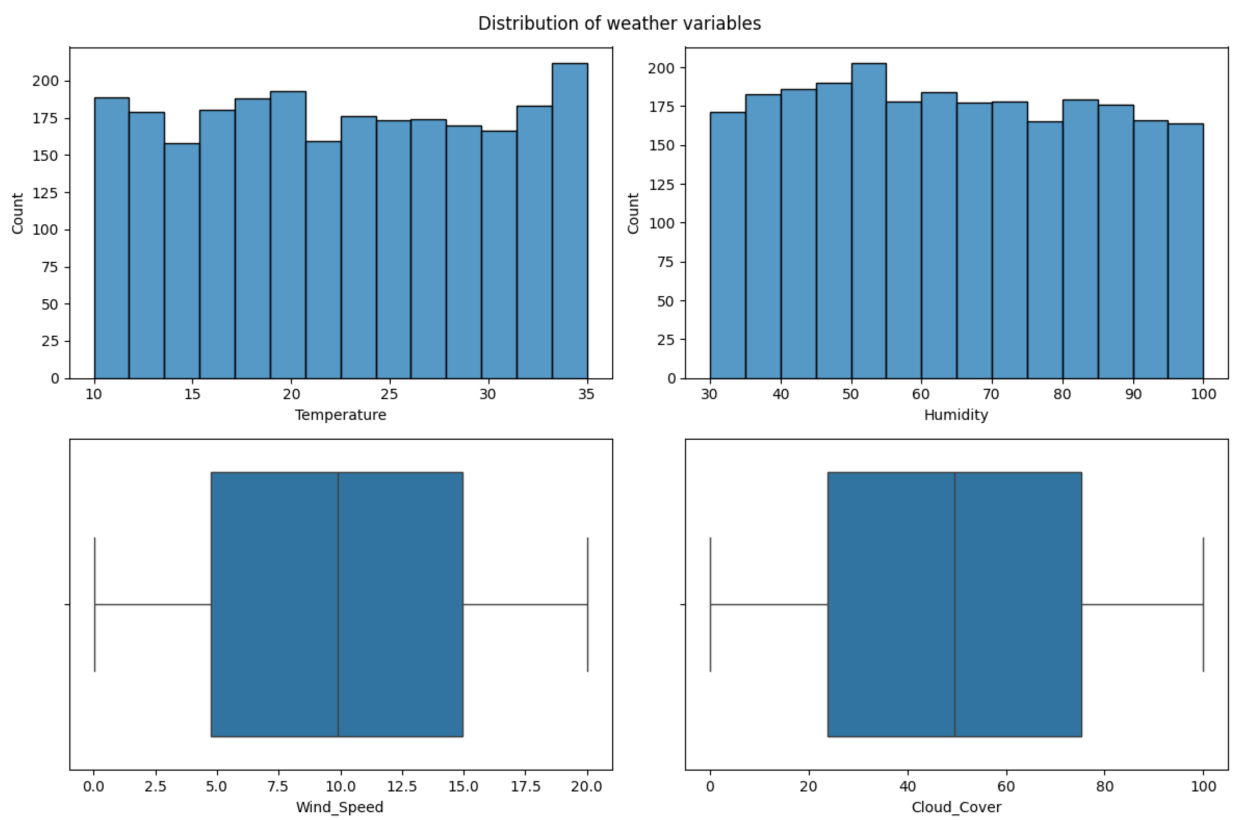
Before Cleaning

Example Data

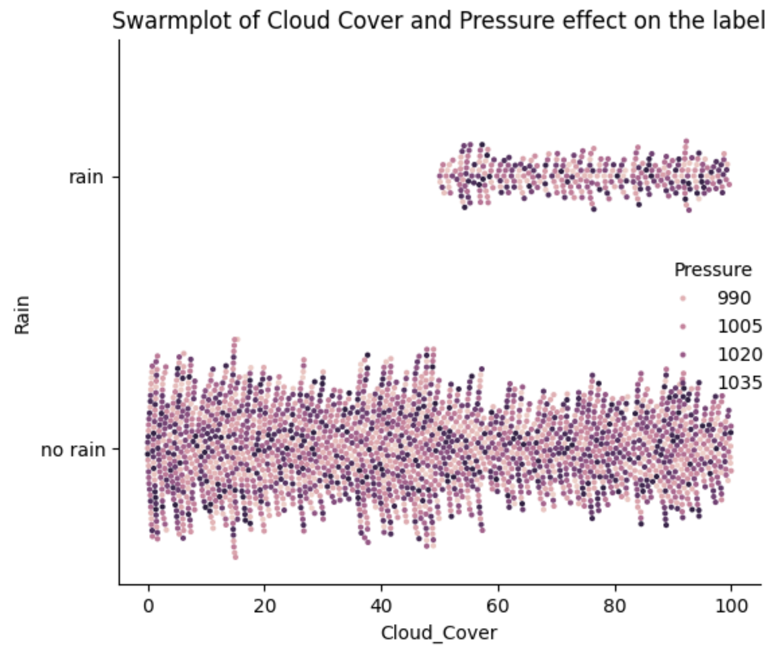
	Temperature	Humidity	Wind_Speed	Cloud_Cover	Pressure	Rain
0	23.720338	89.592641	7.335604	50.501694	1032.378759	rain
1	27.879734	46.489704	5.952484	4.990053	992.614190	no rain
2	25.069084	83.072843	1.371992	14.855784	1007.231620	no rain
3	23.622080	74.367758	7.050551	67.255282	982.632013	rain
4	20.591370	96.858822	4.643921	47.676444	980.825142	no rain

Visualizing Data

The Rain Occurrence dataset is quite curated, especially since it was sourced from Kaggle with the intention of being used as a toy dataset for machine learning applications. Because of this, the dataset is extremely clean right off the bat. For example, below are shown the distributions of the Temperature, Humidity, Wind_Speed, and Cloud_Cover variables. As the reader can no doubt tell, the distributions of each these variables are incredibly uniform and balanced.

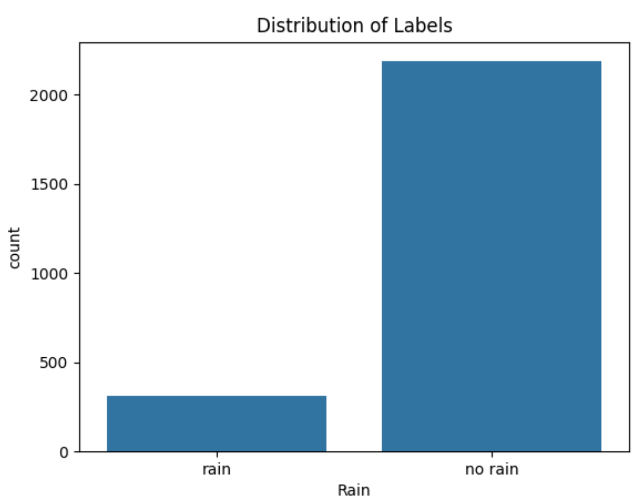


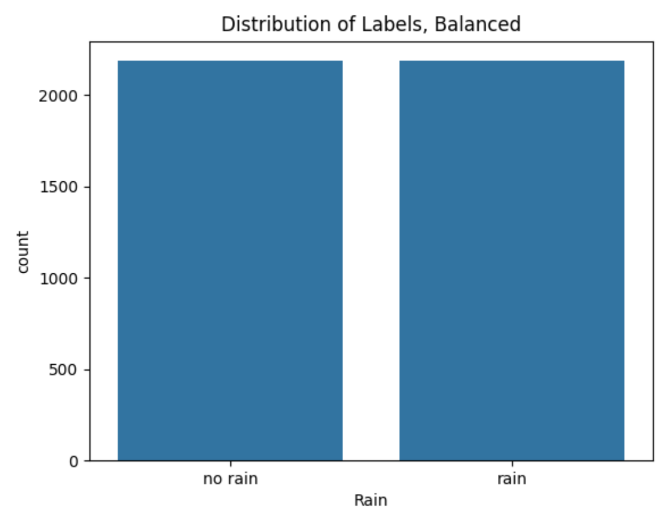
A mixture of histograms and boxplots was used to explore different visualizations in the hope that any obvious outliers would be discovered.



Further exploration uncovered very little in the way of outliers or completely erroneous data points. To the right, a beeswarm plot exploring the relationship between the Rain label and the variable Cloud_Cover can be seen. Examining this plot reveals an obvious correlation between the presence of dense cloud cover and rainfall. This constitutes a wonderful sanity check for the validity of this dataset. If instead the reverse correlation was seen, there may be cause to doubt the accuracy of the data collected.

Interestingly, this plot uncovered a distinct imbalance in the dataset's distribution of the label itself. A simple count plot to the right visualizes this imbalance nicely. Labels indicating a day when it rained comprise only 12.56% of the rows present in the dataset. Using SkLearn's resample utility method, the minority rain label was upsampled, resulting in a 50-50 split. This technique should hopefully limit model bias in later cases.





After Cleaning

Once the dataset was fully cleaned and balanced, there were approximately 4700 rows present. Each variable was either a Float64 representing quantitative data, or the variable was the label column. All data fell into a roughly uniform distribution without statistically significant outliers. The smallest values of data approached 1, while the largest were upwards of 1000. This spread may require some normalization to smooth the use of distance based models such as K-Means or KNN. The final, clean dataset was output to a fresh .csv file.

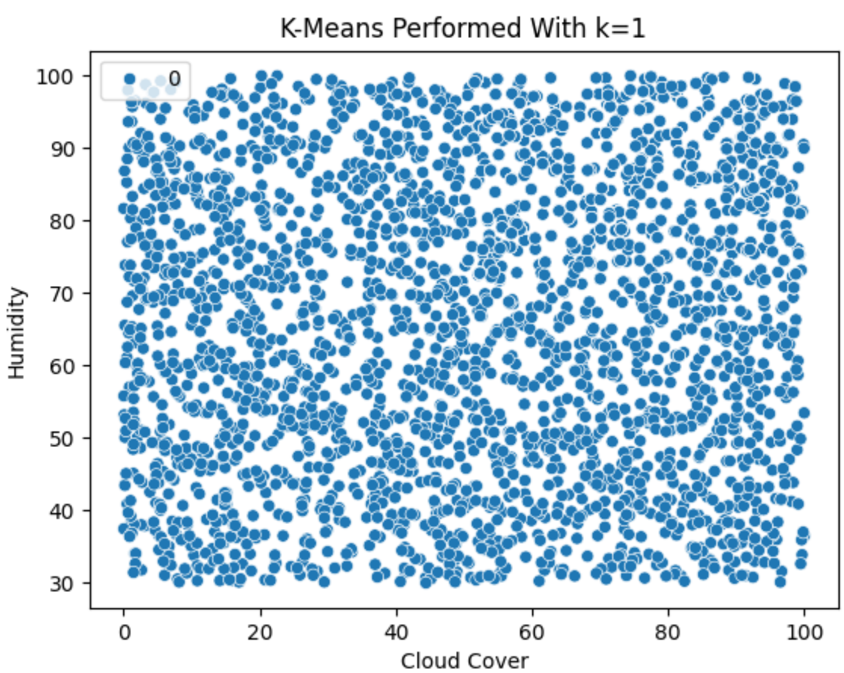
Clustering & PCA

K-Means without PCA

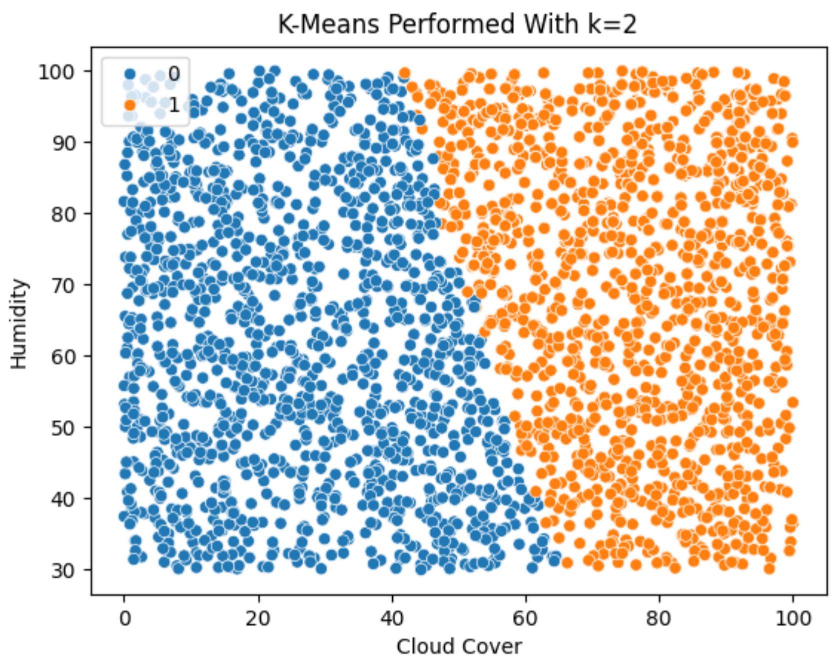
Clean & Prepared Data

	Temperature	Humidity	Wind_Speed	Cloud_Cover	Pressure	Rain
0	27.879734	46.489704	5.952484	4.990053	992.614190	no rain
1	25.069084	83.072843	1.371992	14.855784	1007.231620	no rain
2	20.591370	96.858822	4.643921	47.676444	980.825142	no rain
3	26.147353	48.217260	15.258547	59.766279	1049.738751	no rain
4	20.939680	40.799444	2.232566	45.827508	1014.173766	no rain

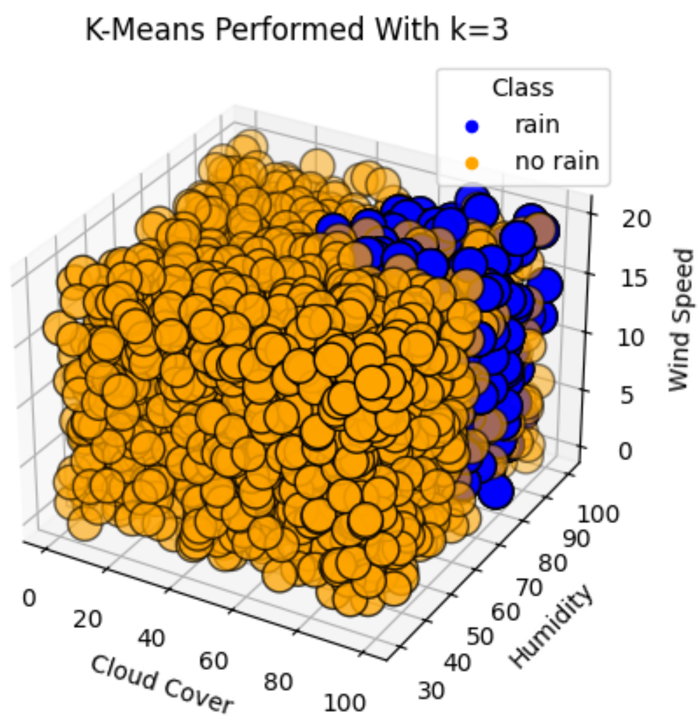
K = 1



K = 2


K-Means Performed With k=2
0
1
Humidity
Cloud Cover
100
90
80
70
60
50
40
30
0
20
40
60
80
100

K = 3


K-Means Performed With k=3
Class
rain
no rain
Wind Speed
20
15
10
5
0
Humidity
100
90
80
70
60
50
40
30
Cloud Cover
0
20
40
60
80
100

Discussion

In this section, unsupervised learning with k-means clustering was performed. As can easily be seen by the plots above, this dataset has an extremely uniform distribution of information. This problem certainly impacts the ability of the model to separate clusters out into distinct and fully disjoint groups. However, as the dimensionality of the visualization increases, the viewer may notice that two clusters begin to appear.

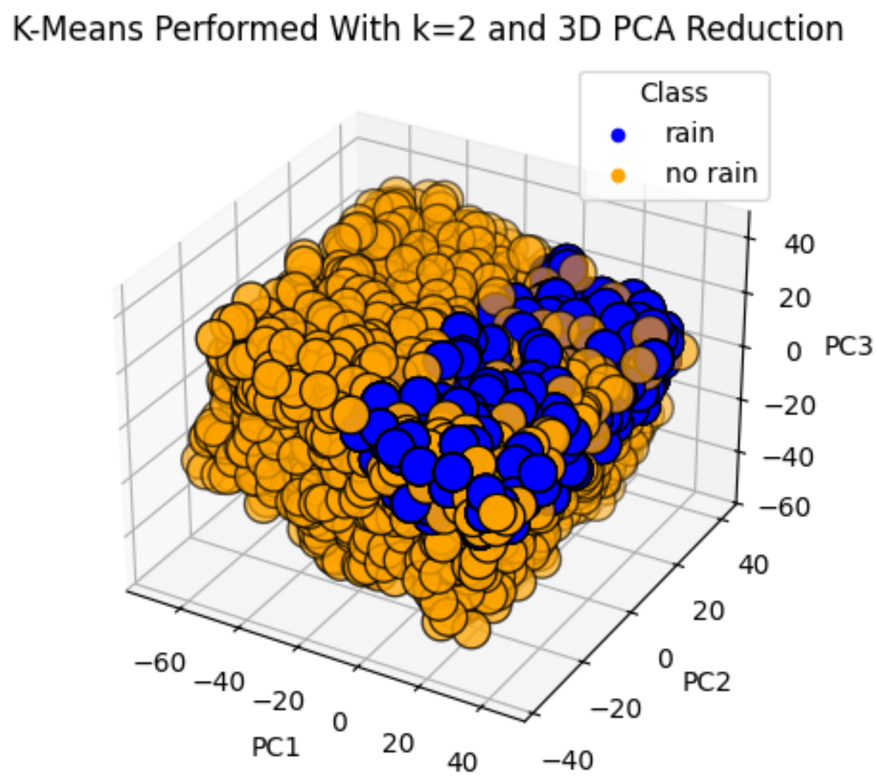
When $k = 1$, the uniformity of the data presents with a single huge cluster spreading to each corner of the plot. As k increases to 2 clusters, a perfect diagonal split appears across the plot of Humidity vs. Cloud Cover. It's worth noting that this form the plot is taking on is not ideal for visualizing clusters, as it only covers 2 of the 5 dimensions in the plot. Finally, with $k = 3$ and a 3D scatter plot of Humidity, Cloud Cover, and Wind Speed, a distinct cluster begins to appear in the top right corner closest to the viewer. It should be noted that in this case, k being equal to 3 has nothing to do with the quality of visualization as there are only two classes present in the original dataset.

K-Means with PCA

Clean & Prepared Data

	PC1	PC2	PC3
0	-62.539052	-16.832413	-11.349318
1	-41.364978	-7.271574	22.050740
2	-7.529130	-36.126166	22.517573
3	-6.096091	38.106757	-19.604017
4	-23.557438	4.065634	-26.199022
...	...	...	...
4367	29.213275	6.703061	-3.636588
4368	24.212369	-29.328186	3.546598
4369	28.854736	-32.301937	1.604066
4370	10.453159	-34.939502	-4.117679
4371	34.508437	6.362508	2.310638

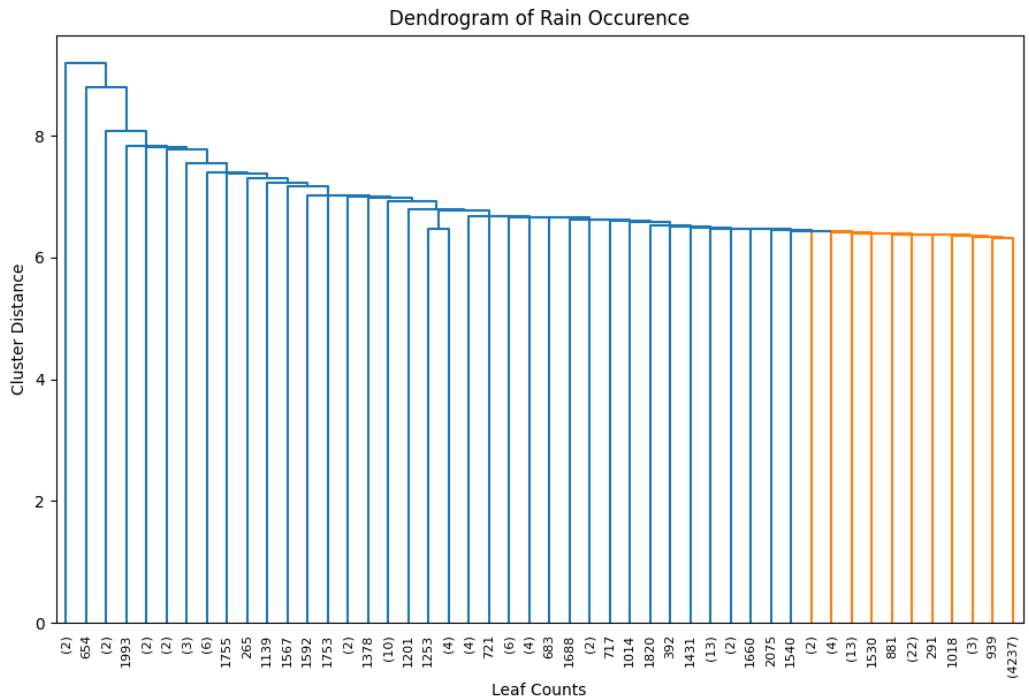
3D PCA Reduction & Clustering



Discussion

Rather unexpectedly, Principle Component Analysis did not have the effect of breaking up the uniformity present in the data. Instead, PCA appeared to have preserved the structure and relationship of each data point to all the others, resulting in the same perfect cube as can be seen in the pure K-Means section above. Interestingly, the use of PCA tilted the cube along the axes corresponding to what must be the direction of each discovered component. All in all, this plot is quite useful. The viewer can easily discern the importance of a high PC1 and PC3 to find whether or not it rained.

Agglomerative Hierarchical Clustering



Discussion

The above plot depicts a dendrogram showing the results of a different clustering technique called Hierarchical Agglomerative Clustering (HAC). Without specifying how many groups should exist within the data, the HAC model discovered the same two groups as before. Unfortunately, the colors of each cluster are reversed in this diagram, so orange corresponds to 'rain' and blue corresponds to 'no rain'. As the viewer can see, the 'rain' cluster still comprises a tight cluster that is highly mixed with the 'no rain' cluster. An interesting thing to note is the one cluster near the middle of the dendrogram with nearly 1300 leaves in it that joins the rest of the 'no rain' cluster late.

As for the differences between k-means and HAC, in this case, very little felt different. Both models discovered two distinct groups in the data, which makes sense since there were two labels in the original dataset, matching it perfectly. Something that was very interesting in both models was the tight grouping of the 'rain' cluster. Part of the process for cleaning and formatting this dataset included up-sampling the 'rain' label, as it had comprised less than 15% of the original data. In the clean dataset, both labels are represented nearly equally. However, clustering of the dataset shows a very distinct range wherein the 'rain' labels should reside, despite the uniform distribution of the data.

Linear Regression Analysis

Preparation & Plan

To work with purely Linear Regression, the Urban AQI Data was used. As mentioned in the gathering and cleaning sections above, this dataset contains a large number of columns labeled by a "Health Risk Score". This score is a continuous value representing the risk of health damage from the air quality of the given city.

```
        tempmax    tempmin       temp  feelslikemax  feelslikemin  feelslike        dew   humidity  ...  solarradiation  solarenergy   uvindex  severerisk  Temp_Range  Heat_Index  Severity_Score  Health_Risk_Score
0    106.100000  91.000000  98.500000    104.000000     88.100000  95.900000  51.500000  21.000000  ...      261.400000    22.500000  9.000000   10.000000   15.100000   95.918703        4.430000          10.522170
1    103.900000  87.000000  95.400000    100.500000     84.700000  92.300000  48.700000  21.500000  ...      293.300000    25.200000  9.000000   10.000000   16.900000   92.281316        3.880000          10.062332
2    105.000000  83.900000  94.700000     99.900000     81.600000  90.600000  41.700000  16.900000  ...      327.000000    28.200000  9.000000   10.000000   21.100000   90.599165        3.630000           9.673387
3    106.100000  81.200000  93.900000    100.600000     79.500000  89.800000  39.100000  15.700000  ...      276.800000    24.000000  9.000000   10.000000   24.900000   89.638811        2.851200           9.411519
4    106.100000  82.100000  94.000000    101.000000     80.000000  90.000000  40.100000  15.900000  ...      274.900000    23.700000  9.000000   10.000000   24.000000   89.760414        3.390800           9.515179
..          ...        ...        ...           ...           ...        ...        ...        ...  ...             ...          ...       ...         ...         ...         ...             ...                ...
993   76.060546  64.359387  69.002142     77.673823     63.510920  67.003338  59.936776  73.189130  ...       71.458332     6.994928  3.121167   10.563084   12.886665   71.837558        1.957318           8.750142
994   68.409198  65.939319  66.567410     68.956722     64.805635  65.992526  59.010257  74.137401  ...      282.751283    22.619979  8.919897    9.838767    2.613629   72.463491        2.537413           9.118198
995   69.756690  65.286919  65.919492     68.158536     63.662942  67.313322  62.024442  84.650482  ...      272.043693    23.830479  9.089547   10.502440    4.598936   67.560060        3.595470           9.880093
996   77.106797  61.481724  68.106569     76.426959     60.901526  68.094309  63.169608  86.860261  ...      271.006885    22.746818  8.908550    9.847929   15.477717   67.930437        3.498942           9.561602
997   90.923080  79.296868  81.636991     94.180423     78.071851  84.987113  73.393045  74.734715  ...      245.190433    20.839690  6.904597   30.395643   11.017871   86.802712        3.040020          10.978044

[998 rows x 25 columns]
```

There were a variety of non-quantitative columns in the clean dataset kept in the cleaning Urban AQI Data section. In this case, to model the data with Linear Regression, only quantitative columns were kept. On the far right of the data show above, the reader may notice the two columns "Severity Score" and "Health Risk Score". Though these two sound similar, "Severity Score" actually refers to the quality of weather. A higher "Severity Score" may coincide with a heat wave or strong storms such as hurricanes.

Using this dataset to predict the health risk posed to city residents by the air quality is of direct interest to this report. As was discussed in the Introduction, the weather affects the lives of people in many ways, some very obvious and some more obscure. In this case, the weather has a clear connection with air quality in urban areas.

Results

To model the data using Linear Regression, an 80/20 train-test-split was used.

Equation:

$$\begin{aligned}\hat{y} = \ 0.6257 &- 0.0083x_1 + 0.0001x_2 - 0.0105x_3 + 0.0115x_4 - 0.0036x_5 + 0.0045x_6 \\ &+ 0.0084x_7 + 0.0232x_8 + 0.0785x_9 + 0.0003x_{10} + 0.0014x_{11} + 0.0264x_{12} \\ &+ 0.0016x_{13} + 0.0003x_{14} - 0.0014x_{15} + 0.0010x_{16} + 0.0082x_{17} - 0.0002x_{18} \\ &- 0.0022x_{19} + 0.0773x_{20} + 0.0016x_{21} + 0.0035x_{22} + 0.0880x_{23} + 0.3018x_{24}\end{aligned}$$

```
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   tempmax          998 non-null    float64
 1   tempmin          998 non-null    float64
 2   temp             998 non-null    float64
 3   feelslikemax     998 non-null    float64
 4   feelslikemin     998 non-null    float64
 5   feelslike        998 non-null    float64
 6   dew              998 non-null    float64
 7   humidity         998 non-null    float64
 8   precip           998 non-null    float64
 9   precipprob       998 non-null    float64
 10  precipcover      998 non-null    float64
 11  windgust         998 non-null    float64
 12  windspeed        998 non-null    float64
 13  winddir          998 non-null    float64
 14  pressure         998 non-null    float64
 15  cloudcover       998 non-null    float64
 16  visibility       998 non-null    float64
 17  solarradiation   998 non-null    float64
 18  solarenergy      998 non-null    float64
 19  uvindex          998 non-null    float64
 20  severerisk       998 non-null    float64
 21  Temp_Range       998 non-null    float64
 22  Heat_Index       998 non-null    float64
 23  Severity_Score   998 non-null    float64
```

Where each of the 24 coefficients above map, in-order, to the variables shown on the left.

To predict on the first row of the test dataset:

$$< 97.200, 75.600, 86.400, 97.800, 75.600, 86.600, 62.500, 48.000,\\ 0.004, 12.900, 4.170, 12.500, 6.700, 107.300, 1011.600, 6.000,\\ 15.000, 261.800, 22.700, 8.000, 10.000, 21.600, 87.908, 2.850 >$$

$$\begin{aligned}\hat{y} = 0.626 &+ (-0.008)(97.200) + (0.000)(75.600) + (-0.010)(86.400) + (0.012)(97.800)\\ &+ (-0.004)(75.600) + (0.005)(86.600) + (0.008)(62.500) + (0.023)(48.000)\\ &+ (0.078)(0.004) + (0.000)(12.900) + (0.001)(4.170) + (0.026)(12.500)\\ &+ (0.002)(6.700) + (0.000)(107.300) + (-0.001)(1011.600) + (0.001)(6.000)\\ &+ (0.008)(15.000) + (-0.000)(261.800) + (-0.002)(22.700) + (0.077)(8.000)\\ &+ (0.002)(10.000) + (0.003)(21.600) + (0.088)(87.908) + (0.302)(2.850)\end{aligned}$$

$$\hat{y} = 10.602$$

The predicted "Health Risk Score" of 10.602 is remarkably close to the true label of 10.907! Overall, the model's accuracy score was 96.80%. This means that out of 100 predictions, users of this model could expect predictions for AQI impact on public health to be accurate roughly 97 times.

Section Conclusion

While the quality of results attained in this section have been exciting, it's worth noting a few caveats before running up to your project manager to deploy the model. Most importantly, this data was only collected for the month of September in 2024, and in 10 specific cities around the U.S.. Refer to Visualizing Variables on the Urban AQI dataset for a list of the cities involved in this analysis. There is much danger to be encountered in Linear Regression when extrapolating analysis results to predictions that are not within the problem space the model was trained on. However, if one is interested in providing fairly accurate Health Risk Score predictions for a fairly specific list of cities and only in the month of September, this model might do a pretty good job!

Logistic Regression Analysis

Preparation & Plan

The Rain Occurrence dataset was chosen as the subject of analysis for logistic regression. As mentioned in the Rain Occurrence data cleaning and preparation section, this dataset consists entirely of quantitative data. Each row is also labeled with a binary classification of either "rain" or "no rain", making it ideal for the use of logistic regression. The cleaned, balanced, and normalized dataset is shown below, coming in at 4372 rows:

	Temperature	Humidity	Wind_Speed	Cloud_Cover	Pressure	Rain
0	27.879734	46.489704	5.952484	4.990053	992.614190	no rain
1	25.069084	83.072843	1.371992	14.855784	1007.231620	no rain
2	20.591370	96.858822	4.643921	47.676444	980.825142	no rain
3	26.147353	48.217260	15.258547	59.766279	1049.738751	no rain
4	20.939680	40.799444	2.232566	45.827508	1014.173766	no rain
...	...	...	...	...	...	...
4367	12.551793	76.913835	0.661807	89.111234	1022.714624	rain
4368	22.918468	87.532690	6.525432	83.528320	987.457506	rain
4369	19.816892	87.193989	12.554171	88.536422	984.560082	rain
4370	23.222373	76.877943	15.825673	72.869790	980.108934	rain
4371	17.478841	84.518188	0.584854	92.551806	1023.389889	rain

[4372 rows x 6 columns]

Using this dataset, the plan is to predict whether or not it was raining at the location and instantaneous time that this data was sampled. Using logistic regression to model this data is a strong choice since each row should be classified into a binary "rain" or "no rain". Practicing building useful and accurate models with a smaller dataset such as this one will allow for the discovery of appropriate methods for modelling larger datasets like the ERA5.

Results

Implementing the Logistic Regression model on this dataset was not terribly difficult. With the labels removed and an 80/20 train-test-split, the data was fit to the model.

General Linear Equation:

$$\hat{y} = -10.730 - 0.274x_1 + 0.153x_2 - 0.007x_3 + 0.091x_4 - 0.001x_5$$

Where x_1 is Temperature, x_2 is Humidity, x_3 is Wind Speed, x_4 is Cloud Cover, and x_5 is Pressure.

To make a new prediction on this dataset, each value in a given data vector can be substituted into its appropriate spot on the linear equation. For instance, using row 3851 of the data rounded to 3 significant figures:

$$< 15.388 \;\; 87.687 \;\; 8.25 \;\; 95.396 \;\; 1029.622 >$$

$$\hat{y} = -10.730 - 0.274(15.388) + 0.153(87.687) - 0.007(8.25) + 0.091(95.396) - 0.001(1029.622)$$

$$\hat{y} = 6.063$$

This linear prediction is then passed through the Sigmoid function to find a value between 0 and 1 representing the probability of rain occurring at the conditions represented by our original vector. The closer a value is to 1, the higher probability that event has, vice versa.

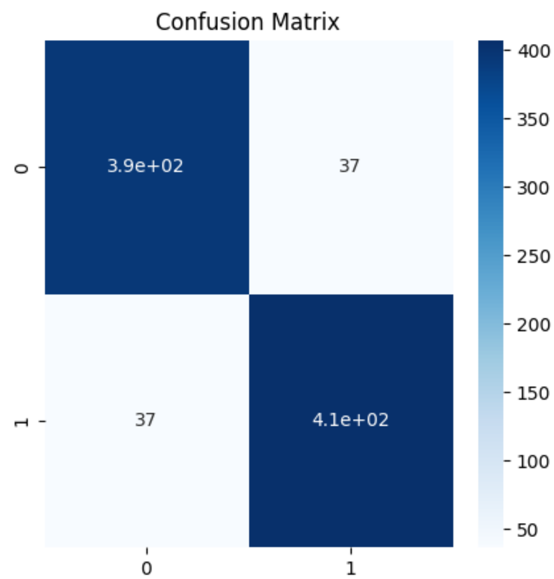
$$\sigma(6.063) = \frac{1}{1 + e^{-6.063}}$$

$$= 0.998$$

The result of 0.998 indicates that the model is very sure that, given the conditions in the first row of the dataset, it was likely raining. Indeed, after accessing the original dataset, that row's label is "rain".

```
Temperature      15.387692
Humidity         87.687334
Wind_Speed        8.252074
Cloud_Cover      95.395951
Pressure       1029.621679
Rain                  rain
```

This is just a single datapoint and the reader may rightfully be skeptical of the model's efficacy. However, on the 20% of data withheld from the training process, this Logistic Regression model predicted with an accuracy of 91.54%. Given the confusion matrix shown to the right, the model in combination with this particular dataset produced a majority of true positives and negatives. Interestingly, exactly the same quantity of false negatives and false positives were predicted! These are cases where the model predicted "rain" when the true label was "not rain" and vice versa.



Section Conclusion

The Logistic Regression model produced above has a fairly strong accuracy score right out of the box. While the Rain Occurrence dataset datatypes are perfectly suited to this kind of classifier, they may not be entirely appropriate. This is because Logistic Regression's foundations are built upon Linear Regression, which has a variety of requirements. For instance, each variable should be linearly associated with the dependent variable, in this case, "rain" or "no rain". Furthermore, each independent variable should have very little to no linear relationship with another variable. If there is some collinearity, one of those variables should either be removed, or a collinear coefficient should be added to the model to account for that relationship. In this case, the former issue of each variable having a linear relationship with the dependent may be in violation. Refer to Visualizing Data in the Rain Occurrence section to see some strong examples of a perfectly uniform distribution of data throughout the problem space. With this in mind, a different supervised model such as Decision Trees or Support Vector Machines could be a better choice for this dataset.

Decision Trees

Preparation & Plan

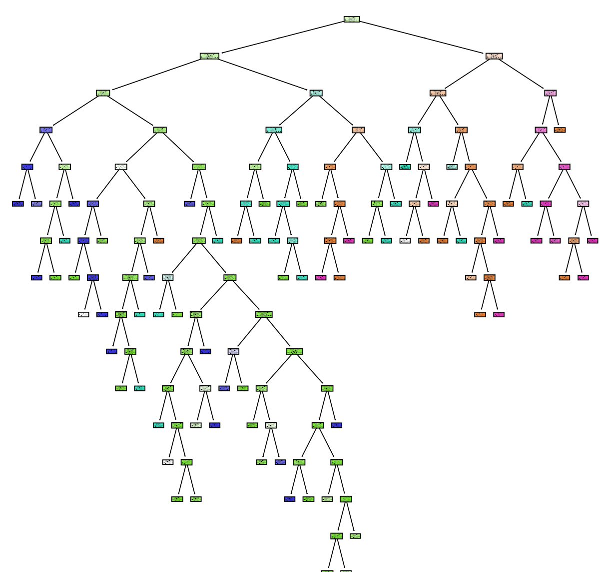
For a demonstration of a Decision Tree Classifier, the Urban AQI Data was chosen. This dataset contains a mixture of qualitative and quantitative data associated with a continuous "Health Risk Score". This score describes the danger to individuals outside in one of the cities contained in the dataset. To prepare the AQI data for Decision Trees, some irrelevant or non-encodable qualitative data was removed, while other qualitative data such as "City" or "Day_of_Week" was ordinally encoded. Most importantly, the "Health Risk Score" was discretized into bins to allow for classification.

	tempmax	tempmin	temp	feelslikemax	feelslikemin	feelslike	dew	...	Temp_Range	Heat_Index	Severity_Score	Month	Day_of_Week	Is_Weekend	health_score_label
0	106.1	91.0	98.5	104.0	88.1	95.9	51.5	...	15.1	95.918703	4.4300	9	2.0	1	High
1	103.9	87.0	95.4	100.5	84.7	92.3	48.7	...	16.9	92.281316	3.8800	9	3.0	1	Medium
2	105.0	83.9	94.7	99.9	81.6	90.6	41.7	...	21.1	90.599165	3.6300	9	1.0	0	Low
3	106.1	81.2	93.9	100.6	79.5	89.8	39.1	...	24.9	89.638811	2.8512	9	5.0	0	Low
4	106.1	82.1	94.0	101.0	80.0	90.0	40.1	...	24.0	89.760414	3.3908	9	6.0	0	Low

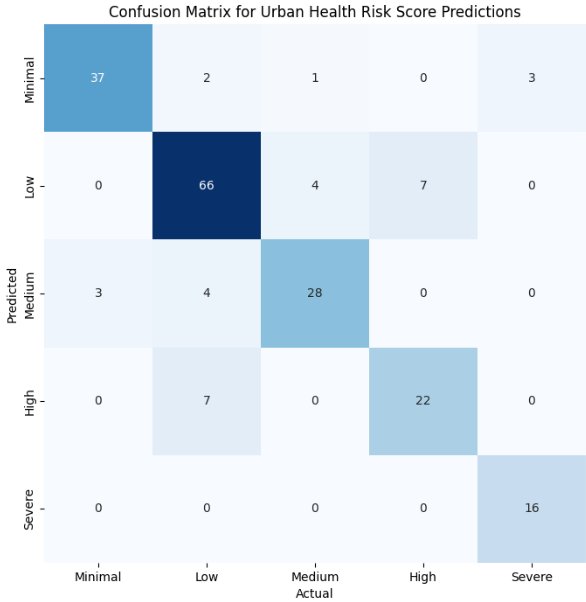
Air quality and the weather are inextricably linked. Finding new ways to predict risk to the public's health due to air quality should be of the utmost importance. Machine learning strategies such as Decision Trees are a great choice for this task. This particular type of model has the unique quality of allowing users to trace the logic of the Decision Tree, understanding why the model came to the decision that it did. Using Decision Trees in this context could help uncover patterns and clues as to how the weather impacts air quality in urban areas.

Results

The first of the two Decision Trees being created by this report can be seen to the right. As the reader can probably tell already, Decision Trees can quickly become quite unwieldly in their size. These models are known for their ability to create arbitrarily convoluted decision boundaries and thus benefit from limiting factors like lowering the maximum depth allowed.



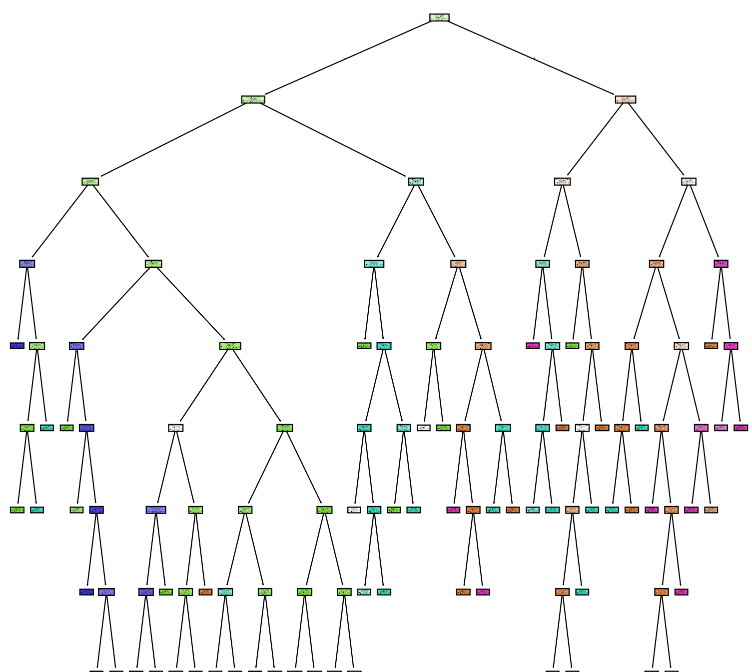
In this case though, an interesting pattern can be seen rather quickly. All the "low" and "medium" risk scores appear to be clustered off the left side of the root, while nearly all the "high" and "severe" labels can be found on the right. This tells us that perhaps more understandable results – and maybe even more accurate modeling – could be achieved with a simpler tree.



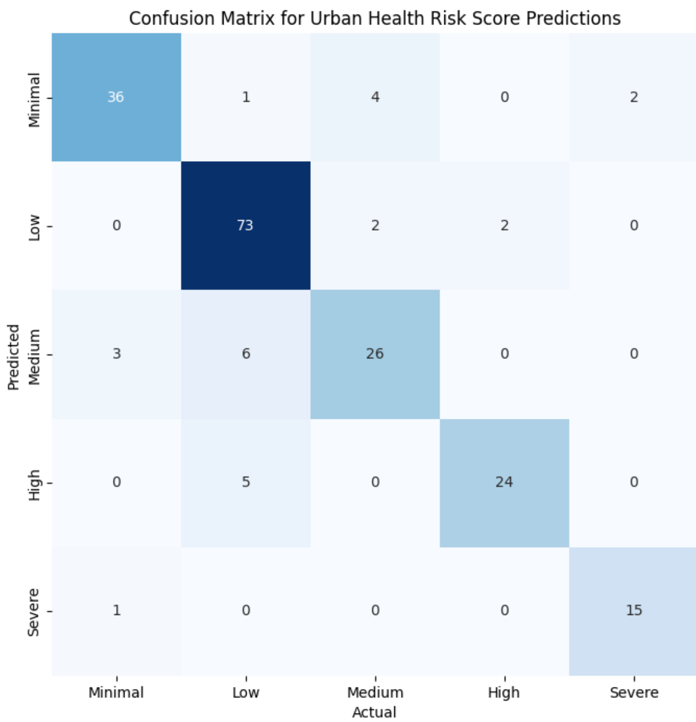

```
------------- Validation -------------
Training Accuracy: 0.845
Max Features: 14
Depth: 15
```

Shown here is validation information regarding the performance of the Decision Tree depicted above. With a maximum number of features limited to 14 and a depth of 15 leaves, this model achieved 84.5% accuracy on the test dataset.

The second decision tree is shown on the right. This tree was limited to roughly half the depth and half the allowed features. As the reader can see, this model is far less complicated than the first. In the worst possible scenario, an analyst would only have to understand 8 decisions for a given prediction as opposed to 15 in the last attempt!



Once again, the same clustering of labels can be seen to the left and right of the root node. This really illustrates how useful decision trees can be. Though not an unsupervised machine learning method, they can still help us understand the structure of our data. Simultaneously, we can ask our decision trees to make predictions of new, unseen data.




```
------------- Validation -------------
Training Accuracy: 0.87
Max Features: 7
Depth: 8
```

Interestingly, limiting the number of features and depth of the new tree resulted in a performance gain of 2.5%! This is most likely due to tradeoffs in overfitting. Given the flexible nature of Decision Trees, it's quite easy for them to become far too specialized in their training data. By limiting the features and depth, users can help the tree generalize to new data more effectively.

Section Conclusion

Predicting a city's air quality health score from a large variety of meteorological factors is not an easy feat, yet a Decision Tree does quite well. Some brief insights can be gleaned from the second – and more useful – model above. For example, the majority of Low and Minimal scoring days depend on a dew point below 63.346, general severity score less than 3.376, and solar radiation levels under 186.792. On the other hand, a city is likely to have a Severe or High scoring day with a dew point above 63.346, and a temperatures above 80.569 degrees.

Naïve Bayes

Preparation & Plan

This section will show an analysis of the Urban AQI Data for both Gaussian and Categorical Naïve Bayes. Each type of Naïve Bayes is suited for different circumstances and data types. Gaussian NB works to classify labels from quantitative data, so predictions on the health risk label will be made from ordinally encoded and otherwise continuous values. Categorical Naïve Bayes makes predictions based on categories, so only ordinally encoded variables will be included for this model.

Data curated for Gaussian Naïve Bayes:

	tempmax	tempmin	temp	feelslikemax	feelslikemin	feelslike	dew	humidity	precip	...
0	106.100000	91.000000	98.500000	104.000000	88.100000	95.900000	51.500000	21.000000	0.000000	...
1	103.900000	87.000000	95.400000	100.500000	84.700000	92.300000	48.700000	21.500000	0.000000	...
2	105.000000	83.900000	94.700000	99.900000	81.600000	90.600000	41.700000	16.900000	0.000000	...
3	106.100000	81.200000	93.900000	100.600000	79.500000	89.800000	39.100000	15.700000	0.012000	...
4	106.100000	82.100000	94.000000	101.000000	80.000000	90.000000	40.100000	15.900000	0.008000	...
..	...	...	...	...	...	...	...	...	...	...
993	76.060546	64.359387	69.002142	77.673823	63.510920	67.003338	59.936776	73.189130	0.084660	...
994	68.409198	65.939319	66.567410	68.956722	64.805635	65.992526	59.010257	74.137401	0.019847	...
995	69.756690	65.286919	65.919492	68.158536	63.662942	67.313322	62.024442	84.650482	-0.010133	...
996	77.106797	61.481724	68.106569	76.426959	60.901526	68.094309	63.169608	86.860261	-0.002440	...
997	90.923080	79.296868	81.636991	94.180423	78.071851	84.987113	73.393045	74.734715	0.151615	...

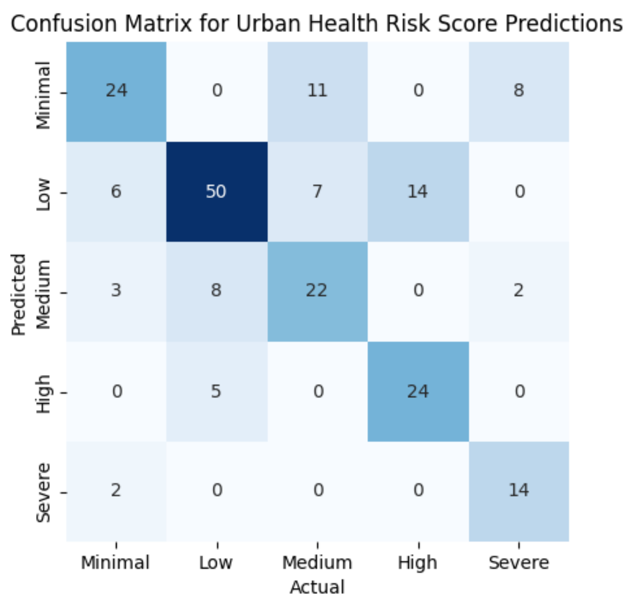
	severerisk	City	Temp_Range	Heat_Index	Severity_Score	Month	Day_of_Week	Is_Weekend	health_score_label
0	10.000000	6.0	15.100000	95.918703	4.430000	9	2.0	1	High
1	10.000000	6.0	16.900000	92.281316	3.880000	9	3.0	1	Medium
2	10.000000	6.0	21.100000	90.599165	3.630000	9	1.0	0	Low
3	10.000000	6.0	24.900000	89.638811	2.851200	9	5.0	0	Low
4	10.000000	6.0	24.000000	89.760414	3.390800	9	6.0	0	Low
..	...	...	...	...	...	...	...	...	...
993	10.563084	5.0	12.886665	71.837558	1.957318	9	6.0	0	Minimal
994	9.838767	8.0	2.613629	72.463491	2.537413	9	5.0	0	Low
995	10.502440	8.0	4.598936	67.560060	3.595470	9	4.0	0	Medium
996	9.847929	3.0	15.477717	67.930437	3.498942	9	2.0	1	Low
997	30.395643	2.0	11.017871	86.802712	3.040020	9	6.0	0	Severe

Data curated for Categorical Naïve Bayes prior to ordinal encoding:

	conditions	City	Month	Day_of_Week	Is_Weekend	Health_Risk_Score
0	Partially cloudy	Phoenix	9	Saturday	True	10.522170
1	Clear	Phoenix	9	Sunday	True	10.062332
2	Clear	Phoenix	9	Monday	False	9.673387
3	Clear	Phoenix	9	Tuesday	False	9.411519
4	Clear	Phoenix	9	Wednesday	False	9.515179
..	...	...	...	...	...	...
993	Partially cloudy	Philadelphia	9	Wednesday	False	8.750142
994	Clear	San Diego	9	Tuesday	False	9.118198
995	Partially cloudy	San Diego	9	Thursday	False	9.880093
996	Clear	Los Angeles	9	Saturday	True	9.561602
997	Clear	Houston	9	Wednesday	False	10.978044

Results

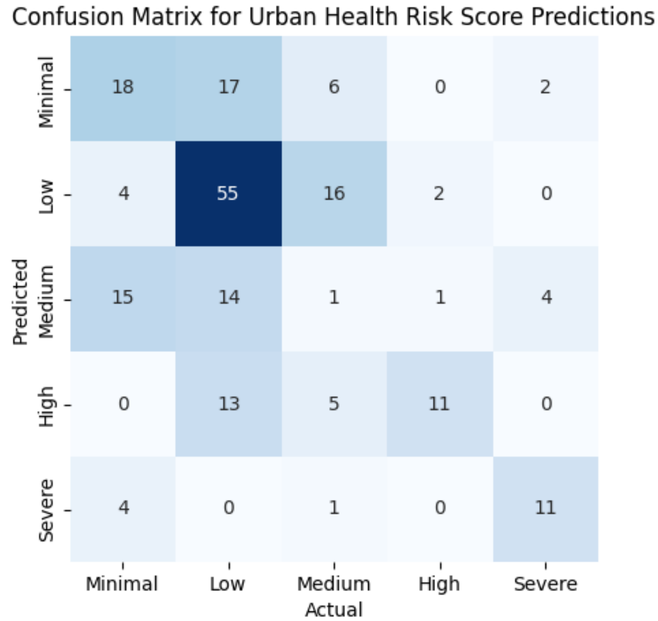
```
Gau Probs:
[[0.978 0.    0.    0.    0.022]
 [0.249 0.    0.    0.    0.751]
 [0.002 0.    0.    0.    0.998]
 [0.    0.    1.    0.    0.   ]
```



The Gaussian Naïve Bayes model's accuracy score came in at a dismal 67%. This is a very low confidence model, and would most likely not be deployed in a professional setting. As can be seen in the confusion matrix on the right, over half of the predictions were correct, aligning along the diagonal. However, a very large quantity of predictions came in as Type 1 and 2 errors. These are False Positives and Negatives.

Above, a small sample of the probabilities for each prediction are shown. The list depicts the probability of each prediction falling into its given category. For the first row, we can see that the model gave this vector a 97.8% probability of having a "minimal" health risk score as well as a 0.22% of a "severe" risk score. Obviously, as human beings, we would expect there to be some middle ground between these predictions. Health risk is not typically a binary question of "minimal" or "severe".

```
Categorical Probs:
[[0.269 0.437 0.264 0.028 0.002]
 [0.688 0.127 0.013 0.007 0.165]
 [0.329 0.015 0.431 0.014 0.21 ]
 [0.149 0.006 0.223 0.002 0.62 ]
 [0.487 0.238 0.118 0.127 0.03 ]
 [0.269 0.437 0.264 0.028 0.002]
 [0.036 0.428 0.272 0.224 0.04 ]
 [0.064 0.174 0.01  0.734 0.019]
```



Incredibly, the Categorical Naïve Bayes model performed even more poorly with a validation accuracy score of 48%, worse odds than 50-50! As can be very clearly seen in the probabilities sample above, there's almost a perfectly even distribution of probabilities across all categories for each row. In the confusion matrix, there appears to be a bias towards predictions of "minimal" or "low", with very few of the predictions going to "high" or "severe", and almost none for "medium".

Section Conclusion

Overall, this demonstration of Naïve Bayes underperformed significantly. However, this is most certainly not due to any intrinsic failings of the model itself. Rather, the nature of this particulary dataset most certainly does not lend itself to successful modeling with Naïve Bayes. That's not to say that the data is unpredictable, as we've seen accuracy ratings of up to 96.80% with Linear Regression Analysis on the same data. The truth is most likely that those variable which were encodable in the Urban AQI dataset were just not *that* significant for predicting label. Given a different dataset with more qualitative, categorical values that are relevant to the prediction space, this model would have performed much better.

Support Vector Machines

Preparation & Plan

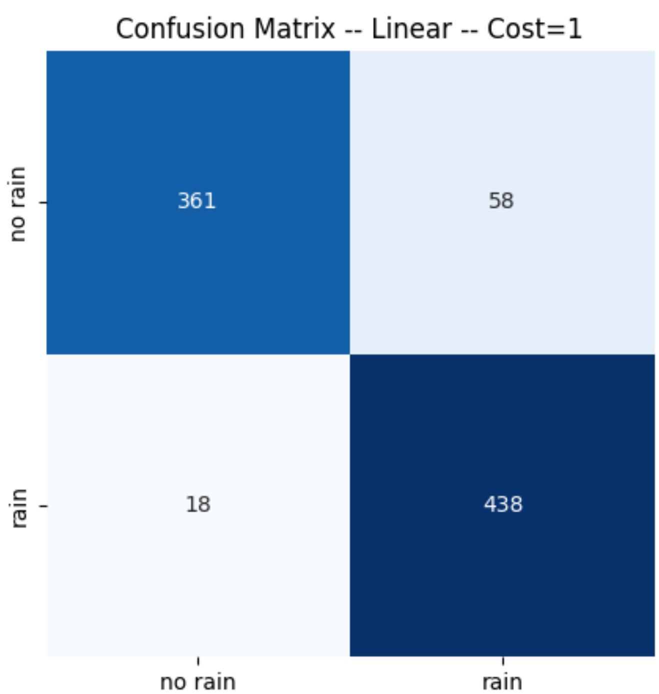
The Rain Occurrence dataset was chosen as the subject of analysis for Support Vector Machine Classification. As mentioned in the Rain Occurrence data cleaning and preparation section, this dataset consists entirely of quantitative data. Each row is also labeled with a binary classification of either "rain" or "no rain", making it ideal for the use of SVM. The cleaned, balanced, and normalized dataset is shown below, coming in at 4372 rows:

	Temperature	Humidity	Wind_Speed	Cloud_Cover	Pressure	Rain
0	27.879734	46.489704	5.952484	4.990053	992.614190	no rain
1	25.069084	83.072843	1.371992	14.855784	1007.231620	no rain
2	20.591370	96.858822	4.643921	47.676444	980.825142	no rain
3	26.147353	48.217260	15.258547	59.766279	1049.738751	no rain
4	20.939680	40.799444	2.232566	45.827508	1014.173766	no rain
...	...	...	...	...	...	...
4367	12.551793	76.913835	0.661807	89.111234	1022.714624	rain
4368	22.918468	87.532690	6.525432	83.528320	987.457506	rain
4369	19.816892	87.193989	12.554171	88.536422	984.560082	rain
4370	23.222373	76.877943	15.825673	72.869790	980.108934	rain
4371	17.478841	84.518188	0.584854	92.551806	1023.389889	rain

[4372 rows x 6 columns]

Using this dataset, the plan is to predict whether or not it was raining at the location and instantaneous time that this data was sampled. SVM should be a great model for this data since each row should be classified into a binary "rain" or "no rain", requiring only the use of one support vector.

Results



To the left, the results of the Support Vector machine modeling can be seen. As predicted, this model was fairly high performing and received an accuracy score of 92.69% on test data. The majority of incorrectly predicted rows were for "rain" when in fact it was not raining.

Using iterative fit-testing the highest scoring models turned out to use a linear kernel with a cost value of 1. Kernels tried included polynomial, sigmoid, linear, and RBF. A range of costs were used as well, from 0.2 up to 8.

Section Conclusion

Of all the models demonstrated in this report up to this point, Support Vector Machines is the only one to have been competitive with Logistic Regression's performance on the Rain Occurrence data set. Perhaps this is because both share the same type of foundation. Each model is a linear classifier and excel at binary classification. The two models perform best when the data is linearly separable, as they output a decision boundary based on a hyperplane. Beyond the most technical matters, such as loss functions and regularization, there are a couple of key differences. While SVMs are excellent at predicting class labels, they are not necessarily meant to predict the probability of that label occurring – something that Logistic Regression is famous for. However, since SVMs approach the problem of separating data based on maximizing the distance from the closest points to the support vector, they are extremely robust to outliers. Given these tradeoffs, it's clear that choosing the correct model depends strongly on the type of data being predicted.

Ensemble Methods

Preparation & Plan

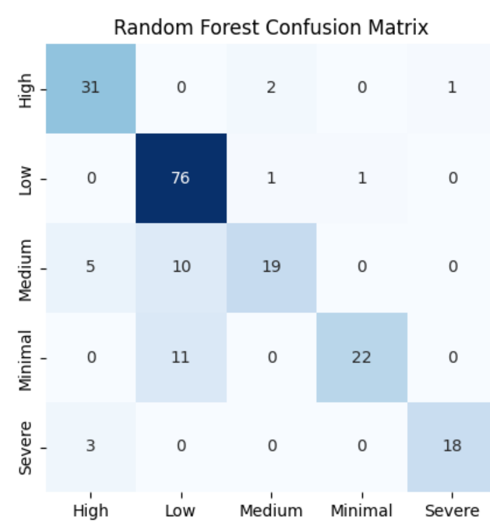
	tempmax	tempmin	temp	feelslikemax	feelslikemin	feelslike	...	Heat_Index	Severity_Score	Month	Day_of_Week	Is_Weekend	health_score_label
0	106.100000	91.000000	98.500000	104.000000	88.100000	95.900000	...	95.918703	4.430000	9	2.0	1	High
1	103.900000	87.000000	95.400000	100.500000	84.700000	92.300000	...	92.281316	3.880000	9	3.0	1	Medium
2	105.000000	83.900000	94.700000	99.900000	81.600000	90.600000	...	90.599165	3.630000	9	1.0	0	Low
3	106.100000	81.200000	93.900000	100.600000	79.500000	89.800000	...	89.638811	2.851200	9	5.0	0	Low
4	106.100000	82.100000	94.000000	101.000000	80.000000	90.000000	...	89.760414	3.390800	9	6.0	0	Low
..	...	...	...	...	...	...	...	...	...	...	...	...	...
993	76.060546	64.359387	69.002142	77.673823	63.510920	67.003338	...	71.837558	1.957318	9	6.0	0	Minimal
994	68.409198	65.939319	66.567410	68.956722	64.805635	65.992526	...	72.463491	2.537413	9	5.0	0	Low
995	69.756690	65.286919	65.919492	68.158536	63.662942	67.313322	...	67.560060	3.595470	9	4.0	0	Medium
996	77.106797	61.481724	68.106569	76.426959	60.901526	68.094309	...	67.930437	3.498942	9	2.0	1	Low
997	90.923080	79.296868	81.636991	94.180423	78.071851	84.987113	...	86.802712	3.040020	9	6.0	0	Severe

For the three Ensemble Methods to be demonstrated in this section – Stacking, Ada Boost, Random Forest – the Urban AQI data set was chosen. This data set is perfect for a shootout between these three common classifying techniques. A majority of the variables in the AQI data are quantitative already, and several are categorical. These variables were ordinally encoded if they were multi-categorical, or binary encoded otherwise. The label of this data, which is a health risk score given to a city based on all the other factors and how dangerous the air may be to people, was binned into 5 categories: minimal, low, medium, high, and severe risk.

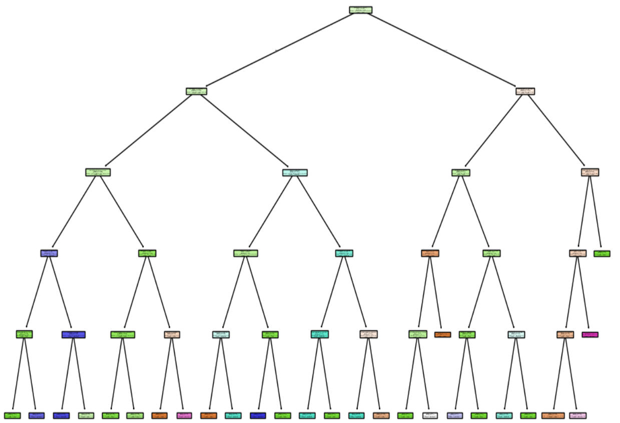
Random Forest

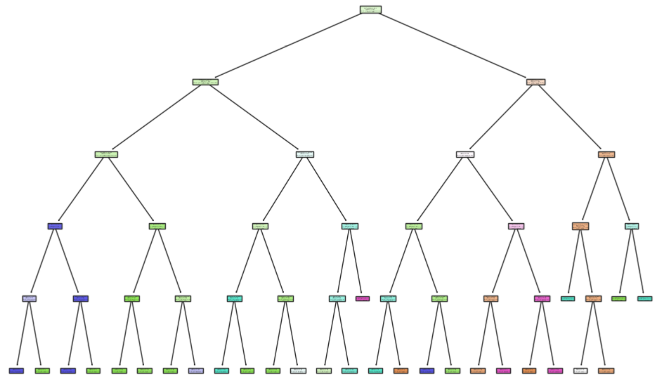
Results & 3 Tree Visualization

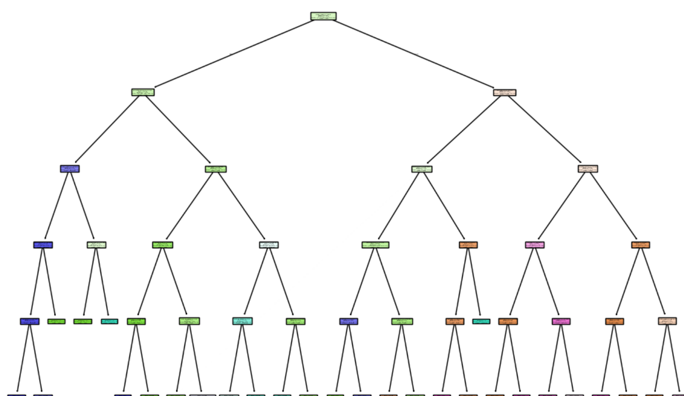
The Random Forest ensemble scored fairly well on the Urban AQI data set with an accuracy right around 83%. Interestingly, the majority of incorrect predictions are clustered in the low category, when they were in fact minimal or medium. This seems to be an acceptable mistake. It would be much worse for the inaccurate predictions to be claiming low risk when actually there is a severe risk to health.



Three of the trees used in this random forest are shown below:

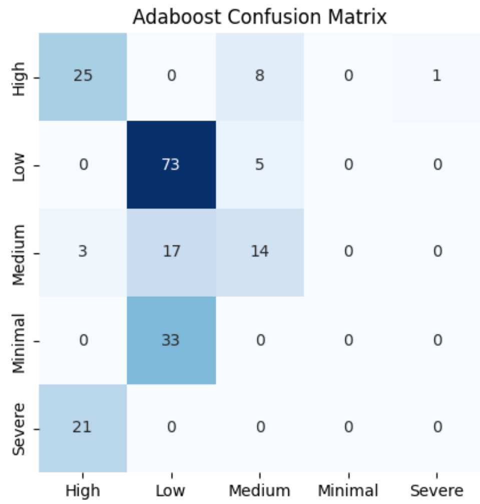






Ada Boost

Results & Ensemble Comparison



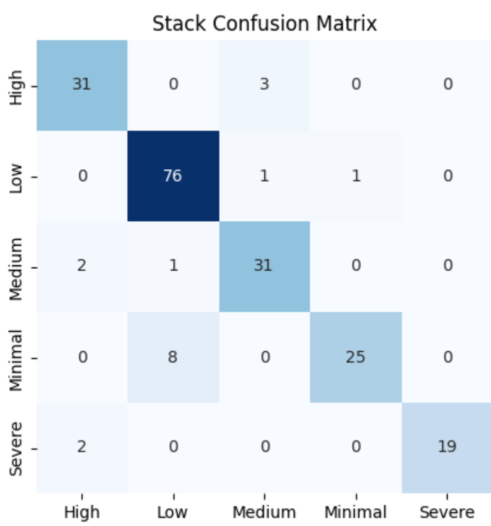
Ada Boost, once again struggles to perform on the given dataset. This model has been troublesome each time it's been implemented. In this case, Ada Boost only scraped a 56% accuracy score on the AQI data set.

Fortunately, the model did not commit the cardinal sin mentioned in the above section. Much can be forgiven of a model which predicts Low risk when it should have predicted Minimal risk. In deployment, if the model were to recommend that people are safe to leave their homes when the AQI is at an all-time high, that could be disastrous.

However, there is still a 27% difference in accuracy between Ada Boost and Random Forest ensemble methods in this case. If a decision was being made between these two on which would be deployed to expensive server space, the Ada Boost ensemble would not be included .

Stacking

Results & Ensemble Comparison



Lastly, the Stack ensemble performed admirably with an accuracy score of 91%. This model was trained using a Random Forest with 10 estimators, a degree 3 polynomial Support Vector Machine with a cost value of 1.0, a Linear Regressor, and a Logistic Regression final estimator.

The ability to customize these ensembles so deeply with so many other types of machine learning models is simply a fantastic thing. Backed by strong domain knowledge, the Stack ensemble can be perfectly tailored to the context in which it will be deployed. This lead to a very strong performance by this technique.

Conclusions

Clustering

In this section, different methods were used to group simple meteorological weather data based on patterns present in the dataset. First, a technique called k-means clustering showed that the data was very evenly spread out, making it hard to form clear groups, although two general patterns did emerge when looking at more features together. Adding a third group didn't actually match the real structure of the data, which originally had just two categories. Then, another method called Principal Component Analysis (PCA) was used to simplify the data, but it didn't break up the even spread, rather, it reshaped the data slightly while keeping the overall structure intact. Lastly, a different grouping method, Hierarchical Agglomerative Clustering (HAC), naturally found the same two groups without needing to be told how many to look for. While the colors representing the groups were flipped in the diagram, the important takeaway was that both k-means and HAC revealed a clear pattern. Even though the data was evenly balanced, the "rain" category stayed tightly grouped together, reflecting a narrow window of weather conditions where rain might occur at the location where this data was gathered.

Linear & Logistic Regression

Linear and logistic regression models are impressive supervised machine learning models for predictive modeling. However, like any model, they do come with important limitations that must be considered or users risk misleading decision makers with their results. Linear regression, for instance, assumes that independent variables are linearly related to the dependent variable and that there is minimal multicollinearity between predictors. Logistic regression, which is fundamentally based on linear regression, inherits many of these assumptions and extends them to classification tasks. When these assumptions are not met, such as when predictors show uniform distributions or nonlinear relationships, model performance degrades.

Importantly, both regression approaches struggle when applied outside the bounds of the training data. Linear regression, in particular, should never be used for extrapolating beyond the specific domain it was trained on. Because of this, even when models produce seemingly strong results on a given dataset, their applicability in broader contexts should be approached with caution. To build robust models, it's essential to pair statistical accuracy with an understanding of data context, model assumptions, and the potential for overgeneralization.

Decision Trees & Naïve Bayes

Decision Trees and Naïve Bayes differ fundamentally in their structure and assumptions, which explains their divergent performance on the air quality prediction task presented to them in this section. A Decision Tree partitions the feature space by learning simple threshold rules, making it great at capturing all kinds of interactions among continuous and discrete variables without requiring any statistical assumptions. Better still, its tree structure allows for interpretability – users can trace exactly which splits lead to a given prediction – a pretty rare quality for a machine learning model.

Naïve Bayes relies on the assumption that all predictors are conditionally independent given the class label, which greatly simplifies probability estimation and generally works well with categorical or discretized inputs. However, as the reader may have noticed in this section, the model can struggle significantly when real-world features exhibit complex dependencies. Naïve Bayes tends toward underfitting when its independence assumption is violated, meaning it can benefit greatly from datasets with a large number of independent categorical variables.

SVMs & Ensemble Methods

Support Vector Machines try to draw the clearest possible line between different groups of data. SVM searches for the best way to separate these groups with a perfectly straight edge while leaving the widest possible gap between the groups. However, if the data isn't neatly separable, they can be bent into curves, stepping into a higher dimension where the separation becomes easier.

When it comes to combining the strengths of multiple models, AdaBoost, Random Forest, and Stacking all take different approaches. Random Forest consist of a large group of trees voting independently on a problem—the idea is that many simple opinions averaged together can be surprisingly accurate. AdaBoost, on the other hand, tries to learn from its mistakes. Each new model tries to correct the errors made by the previous one, gradually improving overall performance. In this section, Ada Boost was not quite able to correct itself, and thus performed the least admirably of the three ensembles. Stacking blends many completely different types of models and uses a final model to make the best possible decision based on all their outputs.

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References

1. Price, I., Sanchez-Gonzalez, A., Alet, F. et al. *Probabilistic weather forecasting with machine learning*. Nature 637, 84–90 (2025). https://doi.org/10.1038/s41586-024-08252-9